

THE NEW SCHOOL • EUGENE LANG COLLEGE OF LIBERAL ARTS

Useful Data

LMTH 2080 — Data and Design with Python

Day 01 • August 31, 2026

Eugene Lang College of Liberal Arts

Key Idea

Data is not useful simply because it exists. It becomes useful in relation to a question, a problem, a person, and a decision.

What Would You Like to Know?

Before we write any Python, begin with a problem or question.

Think about something you would genuinely like to understand, explain, predict, compare, improve, or change. It can be large or small, serious or ordinary.

Data Question

What is a problem or question that interests you?

Now work backward from the question.

problem → **question** → **evidence** → **data**

QUESTION

What would you need to know in order to investigate your question?

DATA

What could you observe, measure, count, record, or collect that would provide evidence?

A Data Hunt

In a small group, choose **one** question to investigate. You have about 20 minutes to find real data on the web that might help.

You are not looking for an article that answers the question. You are looking for the *data* that would allow you to investigate it yourself.

Possible places to look include government open-data portals, research organizations, public APIs, downloadable CSV or spreadsheet files, archives, and datasets published by organizations.

Checkpoint

When you find a promising source, stop and inspect it before continuing to search. The goal is not to collect links. The goal is to understand what you have found.

For your best candidate, answer the following.

1. **What are you trying to understand?**
2. **Where did the data come from?** Who collected or published it?
3. **What does one observation represent?** A person? A trip? A neighborhood? A day? A transaction? Something else?
4. **What variables are available?** What has actually been measured or recorded?
5. **What could you do with this data?** Name at least one question you could plausibly answer.
6. **What is missing?** What would you like to know that this data cannot tell you?

Design Question

Who might find this data useful? What could they understand, decide, or do with it?

Data Is Made

A dataset is not a neutral copy of the world. Someone decided what to count, how to measure it, which categories to use, when to collect it, what to leave out, and how to store it.

Return to the dataset your group found.

LIMITATION

Who or what might be absent from this dataset?

What decisions were made before you ever encountered the file?

If you used this data to make a claim or a decision, what would you want someone else to know about its limitations?

From Data on the Web to Data on Your Computer

Eventually we want to do something with the data we find. That means knowing where our files are and how our projects are organized.

Today we will use the **shell**: a text-based way of communicating with your computer.

Open a terminal. Your instructor will provide a small course folder to work with.

```
pwd
```

`pwd` means **print working directory**. It answers a fundamental question:

Where am I?

Write the result here:

Now ask the computer what is here.

```
ls
```

What do you see?

Moving Around

Directories are folders. Use `cd`—**change directory**—to move into one.

```
cd data
pwd
ls
```

To move back to the parent directory:

```
cd ..
```

Try navigating around the course folder. At any moment, you should be able to answer:

1. Where am I?
2. What files and directories are here?
3. How do I get where I want to go?

Key Idea

You do not need to memorize shell commands today. You do need to begin building a mental model of files, directories, and paths.

Making and Inspecting Things

Try the following commands one at a time. After each command, use `ls` and explain what changed.

```
mkdir scratch
cd scratch
touch notes.txt
ls
cd ..
```

If the course folder contains a CSV file, we can inspect its first few lines without opening a spreadsheet:

```
head data/unemployment.csv
```

A CSV is ultimately a text file. Soon Python will give us much more powerful ways to work with it.

Checkpoint

Challenge: What can you learn from the first few lines?

First inspect the smaller dataset:

```
head data/unemployment.csv
```

Without opening the file in another program, try to answer:

- What does one row represent?
- What variables are available?
- What question could you investigate with this data?

Now try the same thing with a larger dataset:

```
head data/chicago_crimes.csv
```

What changes when a dataset has many more columns?

Is head still enough to understand what you have?

Keeping Track of Change with Git

As we write code, analyze data, and revise projects, our files will change. Git is a system for recording those changes.

Git answers another important question:

What changed?

Move to the Git repository provided for today's exercise and ask Git about its current state.

```
git status
```

Make a small change to the file specified by your instructor. Then run:

```
git status
```

What does Git notice?

To tell Git that a change should be part of our next recorded version:

```
git add <filename>  
git status
```

To record the staged change:

```
git commit -m "Add first notes"
```

And to see the history:

```
git log --oneline
```

Checkpoint

A useful first mental model:

files change → Git notices → we stage changes → we commit a version

GitHub gives a Git repository a remote home. We will use Git and GitHub throughout the semester to organize and share computational work. Branches, merges, and other Git tools can wait.

Before You Leave

Return to the question you wrote at the beginning of class.

QUESTION

What is one question you would genuinely like to investigate this semester?

DATA

What data might help you investigate it?

LIMITATION

What might that data fail to capture?

For Next Time: Find Some Useful Data

Before our next meeting, identify a problem or question that interests you and find **two plausible data sources** that could help you investigate it.

For each source, record:

1. where the data comes from and who produced it;
2. what one observation represents;
3. several important variables;
4. one question the data could help answer; and
5. at least one important limitation or missing piece.

Choose the more promising source and obtain a small sample of the actual data if possible. Bring your question, your sources, and the data to our next meeting.

You will also use the Git workflow from today to add a short Markdown file describing what you found and make your first commit. Detailed submission instructions will be provided with Assignment 0.

Reflection

You are not expected to leave today knowing Unix or Git. You should leave with three habits:

1. begin with a question rather than a tool;
2. ask whether the data you find is actually useful for that question;
3. know where your files are and keep track of how your work changes.